

# DATABASE MANAGEMENT SYSTEM

## Module 2: E-R Modeling & Database Design

Complete Exam Notes with Questions & Answers

### 1. E-R Model – Basic Concepts

#### Q1. What is the Entity-Relationship (E-R) model? Define its basic concepts.

The Entity-Relationship (E-R) model is a high-level conceptual data model used to design a database by representing real-world objects and their relationships in a visual diagram.

It was proposed by Peter Chen in 1976 and is the most widely used technique for database design.

#### Key Basic Concepts:

- **Entity:** A real-world object or thing that has a distinct existence. Example: Student, Employee, Course, Bank.
- **Entity Set:** A collection of all entities of the same type. Example: The set of all Students in a university.
- **Attribute:** A property or characteristic of an entity. Example: Student has attributes – id, name, age, address.
- **Relationship:** An association between two or more entities. Example: Student ENROLLS IN Course.
- **Relationship Set:** A set of all relationships of the same type. Example: All (Student, Course) enrollment pairs.

#### Purpose of E-R Model:

- Helps in planning the database before creating actual tables
- Easy for non-technical users to understand
- Serves as a blueprint for database developers

### 2. Attributes and Entity Sets

#### Q2. What are attributes? Explain all types of attributes with examples.

An attribute is a property that describes an entity. Each entity in the E-R diagram is described by its attributes.

| Type of Attribute   | Definition                                 | Example                                |
|---------------------|--------------------------------------------|----------------------------------------|
| Simple Attribute    | Cannot be divided further; atomic value    | Age, Roll Number                       |
| Composite Attribute | Can be divided into smaller sub-parts      | Name = (First, Middle, Last)           |
| Single-valued       | Holds only one value for an entity         | Date of Birth                          |
| Multi-valued        | Can hold more than one value               | Phone Numbers (a person can have many) |
| Derived Attribute   | Value is calculated from another attribute | Age derived from Date of Birth         |

|                         |                                                        |                         |
|-------------------------|--------------------------------------------------------|-------------------------|
| <b>Stored Attribute</b> | The value from which derived attributes are calculated | Date of Birth           |
| <b>Key Attribute</b>    | Uniquely identifies each entity in an entity set       | Student_ID, Employee_ID |
| <b>Null Attribute</b>   | Attribute with no value (unknown or not applicable)    | Middle Name (optional)  |

#### In E-R Diagram Notation:

- Simple attribute — oval
- Composite attribute — oval with sub-ovals connected
- Multi-valued attribute — double oval
- Derived attribute — dashed oval
- Key attribute — oval with underlined name

### 3. Relationships and Relationship Sets

#### Q3. What is a relationship? What is a relationship set? Give examples.

A relationship is an association among several entities. Example: Student Ravi ENROLLED IN the course DBMS.

A relationship set is a set of relationships of the same type between the same entity sets. Example: {(Ravi, DBMS), (Priya, OS), (Ankit, DBMS)} is the ENROLLED IN relationship set.

#### Degree of a Relationship:

- **Unary (Degree 1):** A relationship between instances of the SAME entity set. Example: Employee MANAGES Employee.
- **Binary (Degree 2):** A relationship between TWO entity sets (most common). Example: Student ENROLLS IN Course.
- **Ternary (Degree 3):** A relationship involving THREE entity sets. Example: Supplier SUPPLIES Part to Project.

#### Descriptive Attributes of a Relationship:

A relationship can also have its own attributes. Example: The ENROLLED IN relationship can have a 'grade' attribute showing the grade a student received in that course.

### 4. Mapping Constraints

#### Q4. What are mapping constraints? Explain cardinality ratios and participation constraints.

Mapping constraints define the number of entities that can be associated with another entity through a relationship. There are two types:

#### A) Cardinality Ratio (Mapping Cardinality)

Expresses the maximum number of entities that can be associated in a relationship.

| Cardinality Type | Explanation & Example |
|------------------|-----------------------|
|------------------|-----------------------|

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| One-to-One (1:1)   | One entity in A is associated with at most one entity in B. Example: One employee manages one department.                        |
| One-to-Many (1:N)  | One entity in A is associated with many entities in B. Example: One teacher teaches many students.                               |
| Many-to-One (N:1)  | Many entities in A are associated with one entity in B. Example: Many students belong to one department.                         |
| Many-to-Many (M:N) | Many entities in A are associated with many entities in B. Example: Students enroll in many courses; courses have many students. |

## B) Participation Constraints

Defines whether ALL or SOME entities participate in a relationship.

- **Total Participation (double line):** Every entity in the entity set MUST participate. Example: Every student MUST enroll in at least one course.
- **Partial Participation (single line):** Some entities may not participate. Example: Not every employee manages a department.

## 5. Keys

### Q5. Define all types of keys used in the E-R model and relational databases.

A key is an attribute (or set of attributes) that is used to uniquely identify an entity in an entity set.

| Key Type             | Definition                                                                | Example                                                       |
|----------------------|---------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Super Key</b>     | A set of one or more attributes that uniquely identify an entity          | <i>{Student_ID}, {Student_ID, Name}, {Email}</i>              |
| <b>Candidate Key</b> | A minimal super key — no unnecessary attributes                           | <i>{Student_ID}, {Email} — both can uniquely identify</i>     |
| <b>Primary Key</b>   | The candidate key chosen by the DBA to uniquely identify rows             | <i>Student_ID is chosen as Primary Key</i>                    |
| <b>Alternate Key</b> | Candidate keys not chosen as the primary key                              | <i>Email is alternate key if Student_ID is primary</i>        |
| <b>Foreign Key</b>   | An attribute in one table that refers to the primary key of another table | <i>dept_id in Student table refers to Department table</i>    |
| <b>Composite Key</b> | A key made of two or more attributes together                             | <i>(Student_ID + Course_ID) in Enrollment table</i>           |
| <b>Partial Key</b>   | Key of a weak entity — unique only with its owner entity's key            | <i>Room_Number alone is not unique; needs Building_ID too</i> |

## 6. Design Issues in E-R Modeling

### Q6. What are the common design issues in E-R modeling?

When designing an E-R diagram, several decisions must be made carefully:

#### Issue 1: Entity vs Attribute

Should something be modeled as an entity or just an attribute?

- **Rule:** If the item has multiple sub-properties or participates in relationships, make it an ENTITY.
- Wrong: Address as a simple attribute of Employee
- Correct: Address as a separate entity if it has sub-parts (street, city, pin code)

#### Issue 2: Entity vs Relationship

Should something be modeled as an entity or a relationship?

- **Rule:** If it connects two or more entities AND has its own attributes, make it an ENTITY (associative entity).
- Example: Enrollment (between Student and Course) has its own attribute 'grade' — so it can be an entity.

#### Issue 3: Binary vs Ternary Relationship

Should a relationship be binary (2 entities) or ternary (3 entities)?

- Example: A doctor prescribes medicine to a patient — this is best modeled as a ternary relationship.

#### Issue 4: Placement of Attributes in Relationships

Attributes of a relationship should be placed on the relationship diamond, not on the entities.

- Example: Grade belongs to the ENROLLED IN relationship, not to Student or Course alone.

---

## 7. Entity-Relationship Diagram (ERD)

### Q7. What is an E-R Diagram? Explain its symbols and how to draw one.

An Entity-Relationship Diagram (ERD) is a visual/graphical representation of the E-R model. It shows entities, their attributes, and the relationships between them.

#### Standard E-R Diagram Symbols:

| Symbol           | What It Represents                              |
|------------------|-------------------------------------------------|
| Rectangle        | Entity Set (e.g., Student, Employee)            |
| Double Rectangle | Weak Entity Set (e.g., Dependent)               |
| Ellipse / Oval   | Simple Attribute (e.g., Name, Age)              |
| Double Ellipse   | Multi-valued Attribute (e.g., Phone Numbers)    |
| Dashed Ellipse   | Derived Attribute (e.g., Age derived from DOB)  |
| Diamond          | Relationship Set (e.g., ENROLLS IN, WORKS FOR)  |
| Double Diamond   | Identifying Relationship (for weak entity sets) |

|                      |                                                      |
|----------------------|------------------------------------------------------|
| Underlined Attribute | Key Attribute (Primary Key)                          |
| Single Line          | Partial Participation                                |
| Double Line          | Total Participation (every entity must participate)  |
| Arrow (→)            | One side of a One-to-One or One-to-Many relationship |
| No Arrow (—)         | Many side of a relationship                          |

### Steps to Draw an E-R Diagram:

- Step 1: Identify all entities (nouns in the problem statement)
- Step 2: Identify attributes for each entity and mark the primary key
- Step 3: Identify all relationships between entities (verbs in the problem)
- Step 4: Determine cardinality (1:1, 1:N, M:N) for each relationship
- Step 5: Identify participation constraints (total or partial)
- Step 6: Check for weak entities and identifying relationships
- Step 7: Draw and finalize the diagram

### Example ERD – University Database:

- Entities: Student, Course, Instructor, Department
- Relationships: Student ENROLLS IN Course (M:N), Instructor TEACHES Course (1:N), Department OFFERS Course (1:N)
- Attributes of Student: student\_id (PK), name, dob, address
- Attributes of ENROLLS IN: grade (relationship attribute)

## 8. Weak Entity Sets

### Q8. What is a weak entity set? How is it different from a strong entity?

A weak entity set is an entity set that does NOT have a primary key of its own. It depends on another entity (called the identifying or owner entity) for its identification.

### Characteristics of Weak Entity:

- Does not have a primary key of its own
- Has a partial key (also called a discriminator) — underlined with a dashed line in ERD
- Must have a total participation in an identifying relationship
- Identified by combining its partial key with the primary key of its owner entity

### Notation in ERD:

- Weak entity — represented by a double rectangle
- Identifying relationship — represented by a double diamond
- Partial key — attribute underlined with a dashed underline

| Strong Entity           | Weak Entity                          |
|-------------------------|--------------------------------------|
| Has its own primary key | Does NOT have its own primary key    |
| Exists independently    | Depends on the owner (strong) entity |

|                                        |                                                    |
|----------------------------------------|----------------------------------------------------|
| Represented by a single rectangle      | Represented by a double rectangle                  |
| Example: Employee                      | Example: Dependent (depends on Employee)           |
| Example: Building                      | Example: Room (depends on Building)                |
| Relationship shown with single diamond | Identifying relationship shown with double diamond |

### Example:

Employee (emp\_id, name) is a strong entity. Dependent (dep\_name, relationship) is a weak entity — dep\_name alone is not unique; it needs emp\_id to be identified. Together (emp\_id + dep\_name) uniquely identifies a dependent.

## 9. Extended E-R Features

### Q9. What are extended E-R features? Explain generalization, specialization, and aggregation.

Extended E-R (EER) features add more expressive power to the basic E-R model to model complex real-world situations. The three main features are:

#### A) Generalization

Generalization is a bottom-up process where multiple lower-level entity sets (subclasses) are combined into a single higher-level entity set (superclass) based on their common attributes.

- Direction: From specific to general (bottom-up)
- Example: Car, Truck, Bus all have common attributes (vehicle\_id, speed, fuel) → generalized into Vehicle
- Used to eliminate redundancy — common attributes are placed in the superclass

*Notation: Triangle symbol labeled 'ISA' (is-a) in the ERD, with subclasses below and superclass above.*

#### B) Specialization

Specialization is a top-down process where a higher-level entity set (superclass) is divided into lower-level entity sets (subclasses) based on distinguishing characteristics.

- Direction: From general to specific (top-down)
- Example: Employee is specialized into Manager, Engineer, Clerk — each has unique additional attributes
- Manager has: department, budget
- Engineer has: project, skills

#### Constraint Types in Specialization:

- **Disjoint:** An entity can belong to ONLY ONE subclass. Notation: 'd' inside circle.
- **Overlapping:** An entity can belong to MORE THAN ONE subclass. Notation: 'o' inside circle.
- **Total:** Every entity in the superclass MUST belong to at least one subclass.
- **Partial:** Some entities in the superclass may not belong to any subclass.

#### C) Aggregation

Aggregation is used when a relationship itself needs to participate in another relationship. It treats a relationship set (along with its entities) as a higher-level entity set.

- Used when a relationship is an entity in another relationship

- Example: Employee WORKS ON Project is a relationship. If we want to add Manager MANAGES (Employee WORKS ON Project), aggregation is used
- The existing relationship (Employee WORKS ON Project) is enclosed in a box and treated as an entity

*Notation: The relationship + its entities are surrounded by a dashed rectangle in the ERD.*

| Generalization                      | Specialization                               |
|-------------------------------------|----------------------------------------------|
| Bottom-up approach                  | Top-down approach                            |
| Combines entities into a superclass | Divides superclass into subclasses           |
| Reduces redundancy                  | Captures distinct sub-type features          |
| Example: Car + Truck → Vehicle      | Example: Employee → Manager, Engineer, Clerk |

## Quick Revision — Key Definitions

| Term                    | Definition                                                                                       |
|-------------------------|--------------------------------------------------------------------------------------------------|
| <b>E-R Model</b>        | A high-level conceptual model to design a database using entities, attributes, and relationships |
| <b>Entity</b>           | A real-world object with a distinct existence. Example: Student, Employee                        |
| <b>Entity Set</b>       | Collection of all entities of the same type. Example: All students in a university               |
| <b>Attribute</b>        | A property describing an entity. Example: name, age, id                                          |
| <b>Relationship</b>     | An association between two or more entities. Example: Student ENROLLS IN Course                  |
| <b>Relationship Set</b> | Set of all relationships of the same type between the same entity sets                           |
| <b>Primary Key</b>      | An attribute (or set) that uniquely identifies each entity in an entity set                      |
| <b>Foreign Key</b>      | An attribute that refers to the primary key of another table — creates a link between tables     |
| <b>Weak Entity</b>      | An entity without its own primary key; depends on an owner/strong entity                         |
| <b>Partial Key</b>      | The discriminator attribute of a weak entity; unique only with the owner's key                   |
| <b>Cardinality</b>      | Number of entities associated in a relationship: 1:1, 1:N, M:N                                   |
| <b>Participation</b>    | Total (all entities involved) or Partial (some entities involved) in a relationship              |
| <b>Generalization</b>   | Bottom-up: combining entities with common features into a higher-level entity (superclass)       |

|                       |                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------|
| <b>Specialization</b> | Top-down: dividing an entity into subclasses based on specific characteristics              |
| <b>Aggregation</b>    | Treating a relationship as an entity to allow it to participate in another relationship     |
| <b>ISA Hierarchy</b>  | The inheritance hierarchy formed by generalization/specialization (subclass ISA superclass) |
| <b>Disjoint</b>       | Specialization constraint — an entity can belong to only ONE subclass                       |
| <b>Overlapping</b>    | Specialization constraint — an entity can belong to MULTIPLE subclasses                     |

---

## Important Exam Questions – Module 2

### Short Answer (2–3 marks):

- Define entity, entity set, attribute, and relationship.
- What is the difference between simple and composite attributes?
- Define primary key, candidate key, and foreign key.
- What is total and partial participation?
- Define weak entity set and give an example.
- What is a partial key (discriminator)?
- Differentiate between 1:1, 1:N, and M:N cardinality with examples.

### Long Answer (5–8 marks):

- Explain all types of attributes with examples and ERD symbols.
- Explain mapping constraints (cardinality + participation) in detail.
- Explain all types of keys used in DBMS with examples.
- What are design issues in E-R modeling? Explain with examples.
- Draw and explain the E-R diagram for a University database / Hospital / Banking system.
- Explain generalization, specialization, and aggregation with examples.
- Differentiate between generalization and specialization.
- What is a weak entity set? How is it represented in an ERD?

— End of Module 2 Notes —